

02 Identification of different celltypes in Seurat

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Introduction

In this RMarkdown the FastQC and multiQC analysis of the .fastq-files from [GSE138852](#) is described. These .fastq-files contain the raw sequencing data and the phred-scores generated on the Illumina NextSeq 500 high-output lane with the 10x Genomics Chromium Single Cell 3' Library & Gel Bead Kit v2. This markdown explains how identification of different cell types is be done and what the conclusion and discussion is in regards to the research question.

Research question

For this project the main research question is *"How can a reliable and reproducible analysis pipeline be established to identify cell-type-specific differential gene expression from snRNA-seq data in Alzheimer's disease?"*, the second step XXXXX

Packages and Requirements

In order to run the scripts for analysis in Seurat a few packages need to be installed first, for analysis an environment was created in miniconda with the command line prompt; XXXXXXXX

Loading the dataset and creating a Seurat object

In order to perform analysis with Seurat, the data first needs to be loaded. This is done with `Read10X()` which is a Seurat function that can be used to load 10X genomics files such as matrices and .tsv files. After loading the dataset, the function `CreateSeuratObject()` is used to create a Seurat object that contains the raw data. In this object several filters can be applied or altered, which is shown in the following sections.

##		orig.ident	nCount_RNA	nFeature_RNA	percent_mt
##	AAACCTGGTAGAAAGG_AD5_AD6	GSE138852	760	564	0.3947368
##	AAACCTGGTAGCGATG_AD5_AD6	GSE138852	720	527	2.9166667
##	AAACCTGTCAAGTCAGT_AD5_AD6	GSE138852	1209	773	1.1579818
##	AAACCTGTCCAAACAC_AD5_AD6	GSE138852	925	615	0.1081081
##	AAACCTGTCCAGTATG_AD5_AD6	GSE138852	562	434	0.5338078

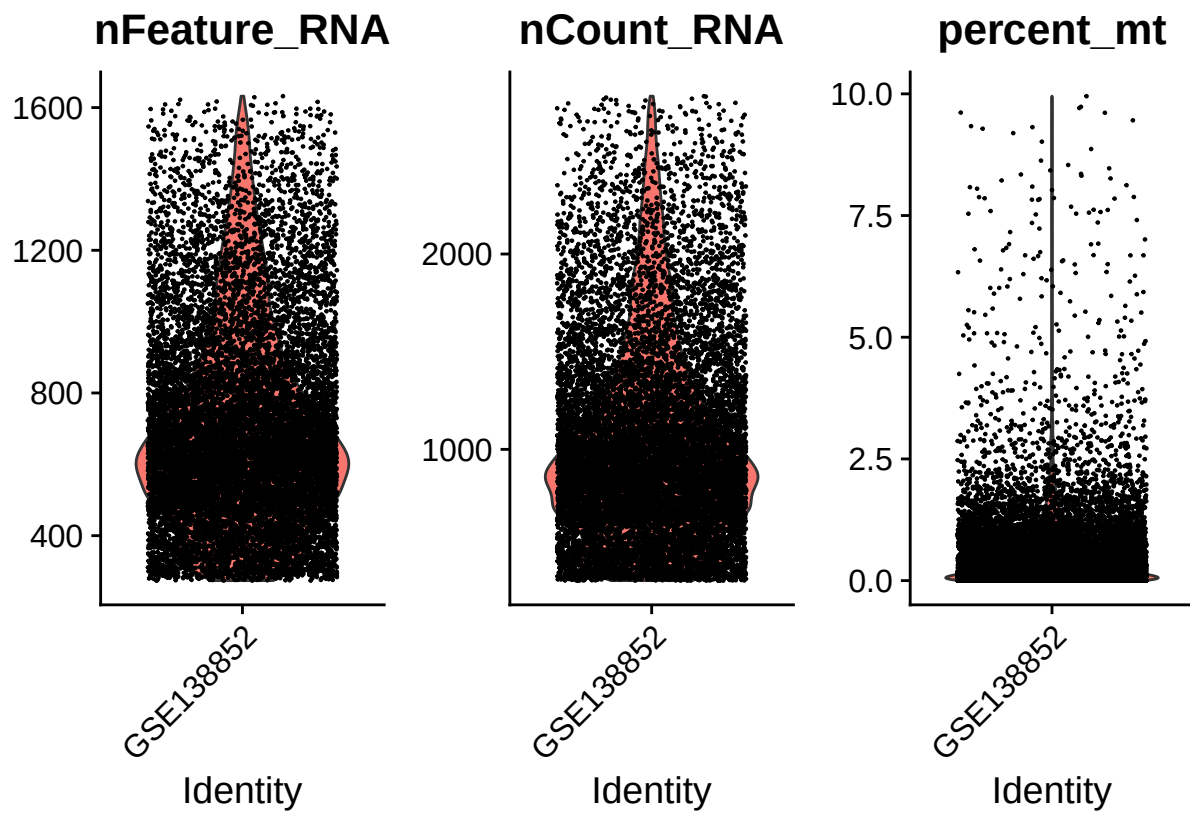


Figure 1: Violin plot of QC-metrics, in the first plot the unique number of genes per cell are visualised, in the second plot the amount of RNA molecules is shown and in the last plot the percentage of mitochondrial genes is shown